

MANIPAL INSTITUTE OF TECHNOLOGY
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Decision Theory and Applied Statistics
Minor-Project Synopsis on

Dynamic Multi-Agent Competition Under Stochastic Demand and Adaptive Learning

SUBMITTED BY

Suryansh Pratap Singh 230929126 (MTE-B)

Akshat Shah 230929130 (MTE-B)

Under the Guidance of:

Prof Sunita Sharma

Department of Mathematics

Manipal Institute of Technology, Manipal, Karnataka (576104) 2025-26

Introduction

Real-world markets operate under uncertainty. Demand changes over time due to trends, seasonality, and random shocks. Firms do not make one-time decisions; they continuously adjust their strategies based on market conditions and competitor actions.

This project builds a dynamic model where demand evolves using a statistically grounded time series model (SARIMAX), and firms compete by adjusting their production decisions over time. The goal is to study how equilibrium forms, whether it is stable, and how demand variability affects this process.

Literature Review

Foundations of non-cooperative equilibrium theory were established by Nash [1], while Cournot [2] introduced strategic quantity competition in oligopolistic markets. Extensions to repeated and dynamic strategic interaction are systematically developed by Fudenberg and Tirole [3]. Stochastic games, incorporating time-dependent payoffs, were introduced by Shapley [4] and further formalized in dynamic settings by Başar and Olsder [5].

Stochastic demand modeling follows the Box–Jenkins framework extended to ARIMAX/SARIMAX models [6], which incorporate exogenous variables alongside autoregressive and moving average components. Econometric foundations for such models, including state-space representations and maximum likelihood estimation, are provided by Hamilton [7]. These models enable the analysis of demand influenced by both internal dynamics and external factors.

Evolutionary stability and adaptive dynamics originate from Maynard Smith [8], with mathematical formulations of replicator dynamics and convergence developed by Hofbauer and Sigmund [9]. Stability analysis of nonlinear dynamic systems follows standard treatments in Strogatz [10].

This project integrates these strands to analyze strategic equilibrium under stochastic and evolving demand conditions.

Problem Statement

The project studies a market with multiple firms where demand evolves stochastically over time. Firms repeatedly update their strategies based on observed outcomes.

Key questions:

- Does equilibrium exist under time-varying demand?
- Do firms converge to equilibrium through adaptive updates?
- How do trend, seasonality, and volatility affect stability?.

Objectives

1. To model market demand using a statistically valid SARIMAX framework
2. To simulate strategic interaction between firms under time-varying demand
3. To analyze whether firms converge to equilibrium through adaptive updates
4. To study the impact of demand trend, seasonality, and volatility on stability

References

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